



ACTIVITY 02 CHECKLIST

Basic Urban Farming with Hydroponics

WSQ Course Code: TGS-2025053916

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Commission, Operate and Restore a Hydroponics Kit

Prepare a kit, verify flow, establish a measured nutrient solution, transplant a seedling and return equipment to a clean serviceable state.

Safety and Scope

- ☐ Correct PPE available
- ☐ Chemical label and SDS available
- ☐ Damaged items quarantined
- ☐ Water and power separated

Detailed Activity Sequence

1. Read the SOP and identify every component and isolation point.

Why: Correct identification prevents misassembly and unsafe intervention.

Acceptance evidence: Component and defect check complete.

2. Inspect, assemble and leak-test before power-on.

Why: Dry checks reduce water, stability and electrical hazards.

Acceptance evidence: Stable leak-free system; cables dry and protected.

3. Power on and verify delivery, return and freeboard.

Why: The farthest site and return establish functional flow.

Acceptance evidence: All sites wet; no overflow or abnormal operation.

4. Measure source-water pH, EC and temperature.

Why: A baseline is required before nutrient correction.

Acceptance evidence: Time, units, meter IDs and calibration status recorded.

5. Mix two-part nutrient in the approved order.

Why: Direct concentrate contact can precipitate nutrients.

Acceptance evidence: Dose and mixing times logged; no precipitate.

6. Verify and incrementally correct EC and pH.

Why: Small measured changes limit overshoot and chemical exposure.

Acceptance evidence: Before/action/after log within trainer setpoints.

7. Accept and place one seedling.

Why: Healthy uniform plugs protect crop consistency.

Acceptance evidence: Label, intact roots, centred support, stable crown.

8. Perform the controlled power-off/restart test.

Why: Commissioning must reveal overflow, siphon and restart risks.

Acceptance evidence: Safe drain-down and restored endpoint flow.

9. Record final readings and obtain trainer verification.

Why: The output state must be objectively verified.

Acceptance evidence: Completed final record and trainer sign-off.

10. Isolate, drain, clean, rinse, dry, inspect and store.

Why: Housekeeping and maintenance prevent contamination and preserve readiness.

Acceptance evidence: Dry complete kit and defect/maintenance record.

Completion Checklist

- ☐ Correct PPE available
- ☐ Chemical label and SDS available
- ☐ Damaged items quarantined
- ☐ Water and power separated
- ☐ Source baseline recorded
- ☐ Nutrients mixed A-then-B into water
- ☐ Final EC/pH within trainer setpoint
- ☐ Transplant acceptance met
- ☐ Power-off/restart safely verified
- ☐ Kit clean, dry, complete and stored

Learner name: _____ Date: _____

Trainer verification: _____